InterPlanetary Atomic Swaps (IPAS): On-Chain Hash Time-Locked Contracts as a Settlement Primitive for the Earth–Mars Economy

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Abstract

Every financial transaction in human history has occurred within one light-speed boundary. Mars breaks that assumption. The 3–22 minute one-way delay, together with two-week solar conjunction blackouts every 26 months, makes centralised exchanges and custodial bridges structurally unviable for Earth–Mars commerce. This paper argues that the correct primitive already exists and has since 2013: the on-chain Hash Time-Locked Contract (HTLC). We call its interplanetary specialisation the *InterPlanetary Atomic Swap* (IPAS). The cryptographic atomicity of an HTLC is distance-agnostic: the hash does not know what planet its preimage was revealed on, and block-height timelocks do not require the two chains' wall clocks to agree. Its operational safety, however, is distance-sensitive: refund windows, monitoring latency, fee buffers, and conjunction handling must be sized against the physics of the link. We present (i) a two-domain systems model in which Earth and Mars maintain locally-validating chains linked only by delayed cross-chain observation; (ii) a timeout-sizing inequality with numerical values for close-, average-, far-, and conjunction-spanning orbital regimes; (iii) the *destination-address construction* that turns a bilateral swap into a three-party commerce primitive without a third trusted role; (iv) a bandwidth analysis showing a Mars settlement can mirror Bitcoin at $\approx$9 MB/hour; and (v) a critical comparison with the Lightning-based proposal of Puente and Puente [2025], arguing that on-chain HTLCs are simpler, more transparent, and better suited to Mars than off-chain channels. The paper is anchored to an existence proof: on 12 April 2026 the Marscoin Foundation completed the first mainnet BTC$\leftrightarrow$MARS atomic swap using 4h/8h timelocks. IPAS extends those parameters — a configuration change, not a protocol change — to 72h/144h at average Earth–Mars distance and multi-week windows across conjunction. For a civilisation in which physical cargo takes months between planets, a six-day financial settlement is effectively instant.

Keywords: atomic swaps, HTLC, Marscoin, Bitcoin, interplanetary commerce, delay-tolerant settlement, Mars economy, solar conjunction, content-addressed settlement.

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1 Introduction

Consider a colonist on Mars who needs a replacement turbopump for a life-support recycler. The supplier is on Earth; the colonist holds Marscoin (MARS); the supplier prices the part in Bitcoin (BTC). A conventional wire transfer requires a correspondent bank that does not exist. A credit-card rail requires an issuer network with merchant-acquirer relationships that do not exist. A centralised cryptocurrency exchange requires a matching engine whose order-acknowledgement loop exceeds the 6–44 minute Earth–Mars round-trip by design. A custodial bridge requires depositing funds with an intermediary 225 million kilometres away, through a communication channel that goes dark for two weeks every 26 months. Every piece of the terrestrial payment stack fails.

What works — what *already* works on mainnet, today, on Earth — is an instrument that was designed for an entirely different purpose and happens, almost accidentally, to be the correct primitive for interplanetary value transfer: the on-chain Hash Time-Locked Contract [Nolan, 2013, Todd, 2014]. An HTLC locks funds on two chains against the same secret hash, with asymmetric refund timers that guarantee either both parties complete or both refund. It does not require synchronous clocks, a trusted custodian, a payment network operator, or continuous connectivity. It requires only that each chain can eventually be observed.

The central claim of this paper is that HTLCs are *accidentally interplanetary.* The properties that make them work for BTC↔LTC swaps in 2017 are the same properties that make them work for BTC↔MARS swaps in 2040. The hash does not degrade with distance. The timelock uses block height, not wall-clock time. The preimage, once revealed on one chain, proves the condition on the other chain in exactly the same way whether the observer detected it in three minutes or twenty-two. The only adaptation required is a parameter change: refund timers extended from hours to days or weeks to absorb light-speed propagation, block-inclusion variance, operational response, and, if needed, conjunction blackout.

We call this construction the *InterPlanetary Atomic Swap*, IPAS. The name deliberately echoes IPFS [Benet, 2014]: just as IPFS retrieves content by hash rather than by server location, IPAS *settles* value by hash rather than by synchronous ledger state. In both cases, cryptography overcomes distance.

An existence proof. On 12 April 2026, the Marscoin Foundation completed the first mainnet atomic swap between Bitcoin and Marscoin using terrestrial-parameter HTLCs [Marscoin Foundation, 2026]. 104 MARS was exchanged for 0.00003652 BTC at the prevailing quote. The swap used a 4-hour BTC timelock and an 8-hour MARS timelock, followed the classical asymmetry rule, and settled without intermediaries. The open-source Electrum-Mars wallet, v4.4.0-mars, drives the full state machine [Lopin, 2026a]. A refund-path test on 11 April — in which a counterparty wallet stalled due to a communication bug, the Bitcoin HTLC was left unfunded on the claim side, and the Bitcoin principal was automatically returned to an external wallet after the 4-hour timelock — independently confirmed that the safety net works even when the happy path fails [Marscoin Foundation, 2026]. IPAS, as described in this paper, is not a new protocol. It is the same protocol, with the timelocks widened to absorb the Earth–Mars link budget. The work is not prospective cryptography; it is parameter selection and operational engineering on a primitive that has been proven in production.

A historical coincidence worth naming. Tier Nolan posted the atomic-swap construction to BitcoinTalk in May 2013. In the same months, a different Bitcoin-adjacent community —

the early Marscoin contributors, the present author among them — was debating how value might move between planets once a Mars settlement had its own chain. One proposal that circulated in those discussions was a *third* blockchain: an "Interplanetary Coin" that would run at a deliberately slower cadence, long enough to tolerate Earth–Mars delay, and that would interface with the native Earth coin on one side and the native Mars coin on the other. The proposal was never fully worked out, and in retrospect the third-chain architecture was the wrong answer to the right question: it introduced a third consensus to maintain, a third token to trust, and an additional attack surface, without resolving the cross-planet settlement problem more cleanly than a bilateral construction could.

Unbeknownst to that discussion, Nolan's atomic-swap construction was already the better mechanism. Two *existing* chains, each sovereign on its own planet, bound together only by the cryptographic condition of a single hash preimage: no third chain, no third token, no third operator. The 2013 Marscoin intuition identified the problem — cross-planet settlement needs to tolerate light-speed delay — and the 2013 HTLC construction supplied the solution, although neither community recognised the fit at the time. Thirteen years later, the pieces connect. IPAS is not a new idea; it is the idea that was implicit in both 2013 conversations, made explicit and deployed.

Contribution. This paper makes five contributions. First, we formalise a two-domain systems model in which Earth and Mars each maintain a locally-validating chain and observe the other only through delayed cross-chain mirrors (Section 3). Second, we present the IPAS protocol itself, including the *destination-address construction* that turns a bilateral swap into a three-party commerce primitive without a third trusted role (Section 4). Third, we derive a timeout-sizing inequality for the Earth–Mars link and supply numerical parameter tables for four orbital regimes (Section 5). Fourth, we bound the bandwidth required for delayed-mirror operation, showing that a Mars settlement can track Bitcoin in approximately 9 MB/hour — trivial for any human-habitable communication link (Section 6). Fifth, we compare IPAS critically with the Lightning-based interplanetary proposal of Puente and Puente [2025], arguing that on-chain HTLCs are strictly simpler, more transparent, more custody-free, and better aligned with the Mars case than off-chain payment channels (Section 7). Sections 8–11 develop the economic and political-economic implications, and Section 12 concludes.

2 Related Work and Historical Context

2.1 HTLCs, atomic swaps, and payment channels

The cryptographic foundation is settled. Nolan [2013] described the HTLC-based cross-chain swap on BitcoinTalk in May 2013. BIP 65 [Todd, 2014] introduced `OP_CHECKLOCKTIMEVERIFY`, the opcode that made HTLCs practical in Bitcoin Script. BIP 112 [Harding and Todd, 2015] added `OP_CHECKSEQUENCEVERIFY` for relative locktimes. The Decred project released the first widely-used implementation in 2017 [Decred Developers, 2017]. Poon and Dryja [2016] generalised HTLCs into routed contingent payments, creating the Lightning Network. Gudgeon et al. [2019] systematised the design space, and COMIT Network [2023] provides the canonical operational statement of the asymmetry rule: *the party who funds second must be able to refund first.* None of this is novel. The IPAS contribution is not the HTLC; it is the deliberate re-targeting of the HTLC to interplanetary parameter ranges.

2.2 Delay-Tolerant Networking and Mars communications

The physical layer is also settled. NASA treats Delay/Disruption Tolerant Networking (DTN) as foundational for a Solar System Internet [NASA, 2023, 2026], with Bundle Protocol Version 7 [Burleigh et al., 2022] and Bundle Protocol Security [Birrane and McKeever, 2022] standardised as IETF RFCs. Earth–Mars one-way light time varies from roughly 3 minutes at closest approach to approximately 22 minutes at greatest separation [NASA, 2020]; solar conjunction blocks direct radio communication for approximately two weeks every 26 months [NASA, 2019]; the relay satellites currently serving surface assets [NASA Science, 2026] already demonstrate store-and-forward operation across the Earth–Mars link. These are the physical parameters into which IPAS must fit, and they are well characterised.

2.3 Blockchain in space

De Filippi and Leiter [2021] survey blockchain-in-space governance proposals, noting that most remain aspirational. Mandl [2017] explored blockchain for distributed spacecraft mission control at NASA. Torky et al. [2022] proposed blockchain authentication for satellite constellations. Our work differs from all three: we do not propose blockchain for mission control or constellation security. We propose a specific cryptographic settlement primitive for interplanetary commerce, anchored in a working mainnet implementation.

2.4 The closest neighbour: interplanetary Lightning

The paper most adjacent to ours is Puente and Puente [2025], *Bitcoin as an Interplanetary Monetary Standard with Proof-of-Transit Timestamping*. Puente and Puente develop a Lightning-centric architecture for Earth–Mars payments, with proof-of-transit receipts, latency-aware CLTV margins, and adapted watchtower logic. Their analysis of light-time-adjusted channel parameters is rigorous, and we adopt several of their sizing intuitions in Section 5.

We differ substantively in two respects. First, we target the *on-chain* HTLC rather than the Lightning channel. Second, we assume *two local monetary domains* (Bitcoin on Earth, Marscoin on Mars) rather than a single Bitcoin monetary standard extended to Mars. Both choices are motivated in detail in Sections 7 and 11, but the short version is: Lightning's terrestrial operational record since 2018 does not inspire confidence that its already-complex operator, channel-balance, and watchtower assumptions will survive 22-minute light delay and 14-day blackouts. The on-chain HTLC has no operator, no channel balance, and no watchtower dependency that cannot be replaced with simple chain observation. For Mars, *simpler is safer.*

2.5 Mars political economy

Zubrin [1995] established interplanetary commerce as central to Mars colonisation economics. The Marscoin white paper [Lopin et al., 2014] argued from first principles that a Mars settlement would need a local chain because light delay makes Earth-bound terrestrial consensus unviable. The Martian Republic paper [Lopin, 2022] extended that logic into governance and IPFS-linked civic infrastructure. Haqq-Misra [2024] argues from a sovereignty angle that a mature Mars may need significant monetary separation from Earth. IPAS is politically neutral among these positions: it provides a mechanism for selective, trust-minimised cross-domain transfer that is compatible with full autonomy, pegged federation, or restricted interchange regimes. What the technology makes possible is a political choice.

3 System Model

3.1 Two monetary domains, not one synchronous ledger

We assume two independent monetary domains:

- **Earth domain** E, with chain C_E (Bitcoin in our running example);
- **Mars domain** M, with chain C_M (Marscoin in our running example).

Each chain validates locally. Neither chain depends on the other for its own consensus. Nodes in domain E may observe blocks of C_M only after propagation delay and relay scheduling; the reverse holds for M. The architecture is *locally validating, cross-domain observing.*

This is a deliberate design choice, not a limitation. A naive design that attempted to run a single base-layer chain across both planets would inherit the worst features of interplanetary latency in every routine transaction. Marscoin's original 2014 insight was exactly this: let each planet clear its own everyday economic life locally, and treat cross-planet settlement as a distinct, less frequent operation [Lopin et al., 2014].

3.2 Actors

IPAS involves two swap participants. Unlike typical bilateral swap presentations, *neither participant is required to be the ultimate beneficiary*:

- **Maker** (P_M): the party who initiates the swap, typically by generating the preimage and funding the first HTLC. In our commerce scenarios, the maker is usually on Mars offering MARS.
- **Taker** (P_E): the party who accepts the offer and funds the counter-HTLC on the other chain.

Each HTLC specifies a *claim destination* — a public key that may or may not belong to the swap participant. The claim destination can be a supplier, a family member, a governance treasury, an institution. This is the *destination-address construction*, which we develop formally in Section 4. It is why IPAS is a commerce primitive, not merely a trading primitive.

3.3 Technical assumptions

We assume:

1. Both C_E and C_M support hash-locked and time-locked script (native for Bitcoin and Litecoin-derived chains including Marscoin).
2. Cross-domain observation is *delayed but eventually consistent* under honest relay.
3. Relay infrastructure may use DTN-style store-and-forward transport.
4. Participants may run automated monitoring workers but are not assumed to operate proprietary watchtower networks.
5. No wall-clock synchronisation between planets is required; each chain advances by its own block height.

We do *not* assume: a single globally synchronised clock, instant mempool visibility across planets, common legal jurisdiction, or uninterrupted communications.

3.4 Threat model

The relevant threats are operational rather than cryptanalytic. Preimage resistance and signature unforgeability are inherited from the underlying chains. The threats IPAS must survive are:

- counterparty abort after one leg is funded;

- relay delay, reordering, or suppression;
- censorship at the network, miner, or relay-operator level;
- local chain reorganisations of lightly-confirmed transactions;
- fee spikes delaying inclusion of claim transactions;
- solar conjunction suspending usable Earth–Mars communication for weeks.

4 The IPAS Protocol

4.1 An HTLC primer for the space audience

The construction is small enough to state precisely. Let $H(\cdot)$ be a cryptographic hash function (SHA-256 in practice) and let s be a uniformly random 32-byte *preimage*. Let $h = H(s)$ be the *hash lock*. An HTLC on chain C locks an amount v such that it can be spent in one of two ways:

$$\mathrm{HTLC}(C, v, h, T, pk_{\mathrm{claim}}, pk_{\mathrm{refund}}) := \begin{cases} \text{claim:} & \text{signed by } pk_{\mathrm{claim}} \text{ AND } s \text{ with } H(s) = h \\ \text{refund:} & \text{signed by } pk_{\mathrm{refund}} \text{ AND current height} \geq T \end{cases} \tag{1}$$

The claim path can be taken at any time, provided the claimer knows s. The refund path can be taken only after the chain's block height reaches T. Two properties matter for what follows:

1. **The hash lock is atemporal.** $H(s) = h$ is a property of s, not of when or where s was revealed. A preimage broadcast on Mars at time t reaches Earth's Bitcoin network at $t + \tau_{\mathrm{prop}}$, and verifies identically and instantly at every node that eventually receives it. The hash function does not know that τ_{prop} is 22 minutes instead of 22 milliseconds.
2. **The time lock uses block height, not wall-clock time.** `OP_CHECKLOCKTIMEVERIFY` compares the transaction's `nLockTime` against the embedded T, both expressed as block heights on the local chain. Earth's clock and Mars's clock need not agree. Each chain advances at its own pace; the refund becomes available when *that* chain's own tip reaches T.

These two properties mean an HTLC constructed for a 4-hour settlement window works identically whether the counterparties are in the same room or on different planets. The only required adaptation is choosing T appropriately.

4.2 The swap

Let Alice be on Mars and hold MARS. Let Bob be on Earth and hold BTC. Alice wants to trade MARS for BTC directed to an address A_B on Earth, which may be her own or a supplier's. Bob wants to acquire MARS directed to an address A_M on Mars, which may be his own or a beneficiary's.

The protocol is identical to a terrestrial atomic swap, up to parameter choice:

1. **Setup.** Alice generates s, computes $h = H(s)$, and shares h with Bob over any channel.
2. **Mars leg.** Alice funds $\mathrm{HTLC}_M = \mathrm{HTLC}(C_M, v_M, h, T_M, pk_B, pk_A)$ on Marscoin. Bob can claim v_M on Mars if he reveals s; Alice can refund after block height T_M on C_M.
3. **Verification.** Bob observes HTLC_M on his local (delayed) copy of C_M and verifies its parameters.
4. **Earth leg.** Bob funds $\mathrm{HTLC}_E = \mathrm{HTLC}(C_E, v_E, h, T_E, pk_{A_B}, pk_B)$ on Bitcoin. Alice (or her designated claimant on Earth) can claim v_E if she reveals s; Bob can refund after block height T_E on C_E.

5. **Claim on Earth.** The Earth-side claimant reveals s and takes v_E to A_B.
6. **Claim on Mars.** Bob observes s on his delayed mirror of C_E, reconstructs the claim transaction on C_M, and takes v_M to A_M.

If Alice never broadcasts HTLC_M, nothing happens and Bob loses nothing. If HTLC_M is funded but Bob never funds HTLC_E, Alice refunds after T_M and loses only time. If both legs fund and no claim is ever made, both refund and both lose only time. If the Earth claim is made, s becomes public and Bob can always claim on Mars, provided $T_E < T_M$ with an adequate safety gap (Section 5).

4.3 The destination-address construction

Equation 1 specifies that the claim path requires a signature by pk_{claim}. That public key does *not* have to belong to the swap participant. It can belong to anyone the participant designates. This is the single insight that transforms an atomic swap from a trading primitive into a commerce primitive.

Consider six worked examples.

Example 1: A turbopump. A colonist on Mars needs a replacement turbopump manufactured by an Earth supplier. The colonist creates an IPAS offer with $A_B = pk_{\text{supplier}}$. An Earth-based Marscoin investor takes the offer, funding the BTC leg. When the swap completes, the supplier receives BTC directly; the colonist's MARS went to the investor; the investor acquired MARS exposure. Three parties, two chains, two planets, zero intermediaries.

Example 2: A birthday. An Earth-based grandparent wants to send birthday money to a grandchild born on Mars. The grandparent takes a MARS offer, specifying the grandchild's Marscoin address as A_M. The grandparent's BTC goes to the maker; the MARS goes to the child. Hawala networks have performed exactly this function for 1,200 years using trusted intermediaries [Zubrin, 1995]. IPAS performs it with mathematics.

Example 3: Scientific data. A Mars colony produces a unique isotopic dataset. A Caltech lab wants it. The lab takes an IPAS offer against the colony's Mars-side treasury wallet. On claim, the BTC lands in the colony's Earth-side logistics account; the MARS the lab acquired funds its cross-planet research retainer. The data itself, content-addressed by IPFS hash, is released on preimage reveal — the *settlement* and the *delivery* share the same cryptographic condition.

Example 4: Emergency medicine. A rare antibiotic manufactured only on Earth is needed urgently on Mars. The Mars habitat uses IPAS to settle payment to the pharma supplier in under a week. Starship cargo delivers the physical goods over the next synodic window. The financial settlement beats the physical shipment by months. The architecture matches the physics.

Example 5: Sovereign treasury funding. A Martian Republic governance treasury issues an IPAS offer of 10,000 MARS at the prevailing quote. An Earth-based impact investor takes it. Capital flows to the Martian Republic within the IPAS settlement window — days, not the years of an institutional grant cycle, and without a NASA or ESA contracting vehicle. The receiving address is the on-chain treasury multisig; the auditability is public.

Example 6: Intellectual property. A Mars-based inventor sells a patent licence to an Earth corporation. BTC lands with the Earth-side patent-firm trust account (the A_B specified in the offer); MARS settles to the inventor's wallet on Mars. The execution of the licence depends on the preimage reveal, which is publicly auditable and timestamped on both chains.

The HTLC itself is indifferent to every one of these. It enforces only: preimage binds both sides, timelocks enforce refunds. What humans choose to do with those mechanics — pay suppliers, support family, fund research, send medicine, capitalise a republic, license a patent — is their sovereign decision.

4.4 Offer propagation

IPAS offers propagate between planets through whatever gossip infrastructure the Marscoin and Bitcoin ecosystems already use. In the current Marscoin deployment, ElectrumX servers gossip swap offers among peers; for IPAS, this gossip traverses the interplanetary link like any other data. The offer format itself requires no protocol change. A future enhancement might add an `ipas: true` flag so wallet UIs can display conjunction warnings and orbital-delay estimates.

4.5 The protocol in a single diagram

Figure 1 summarises the full IPAS exchange as a message-sequence diagram between Mars, the interplanetary link, and Earth. The dashed arrows are cross-planet messages that traverse the light-speed delay τ_{prop}. The solid arrows are local chain transactions. The only global synchronisation required is the publication of h; everything else is driven by each party's independent observation of its local and remote chains.

5 Timeout Sizing

5.1 The asymmetry rule

The canonical atomic-swap rule [COMIT Network, 2023] is:

$$T_E < T_M, \tag{2}$$

where T_E is Bob's refund height on C_E (Bob is the second funder) and T_M is Alice's refund height on C_M. The party who funds second must be able to refund first. Were the inequality reversed, Alice could observe Bob's claim on Earth, then refund her Mars leg before Bob could relay s back to C_M. The rule is not a convention; it is a correctness condition.

Let Δ denote the safety gap:

$$\Delta := \text{wallclock}(T_M) - \text{wallclock}(T_E), \tag{3}$$

that is, the wall-clock time between the Earth refund becoming active and the Mars refund becoming active. Δ is the window during which Bob can observe s on C_E, relay it to his Mars worker, and have the Mars claim transaction confirmed on C_M before Alice's refund race opens.

5.2 The sizing inequality

We decompose Δ into physical, operational, and policy components:

$$\Delta \;\geq\; \tau_{\text{confirm},E} + \tau_{\text{prop}} + \tau_{\text{relay}} + \tau_{\text{obs}} + \tau_{\text{claim},M} + \tau_{\text{confirm},M} + \tau_{\text{fee}} + \varepsilon, \tag{4}$$

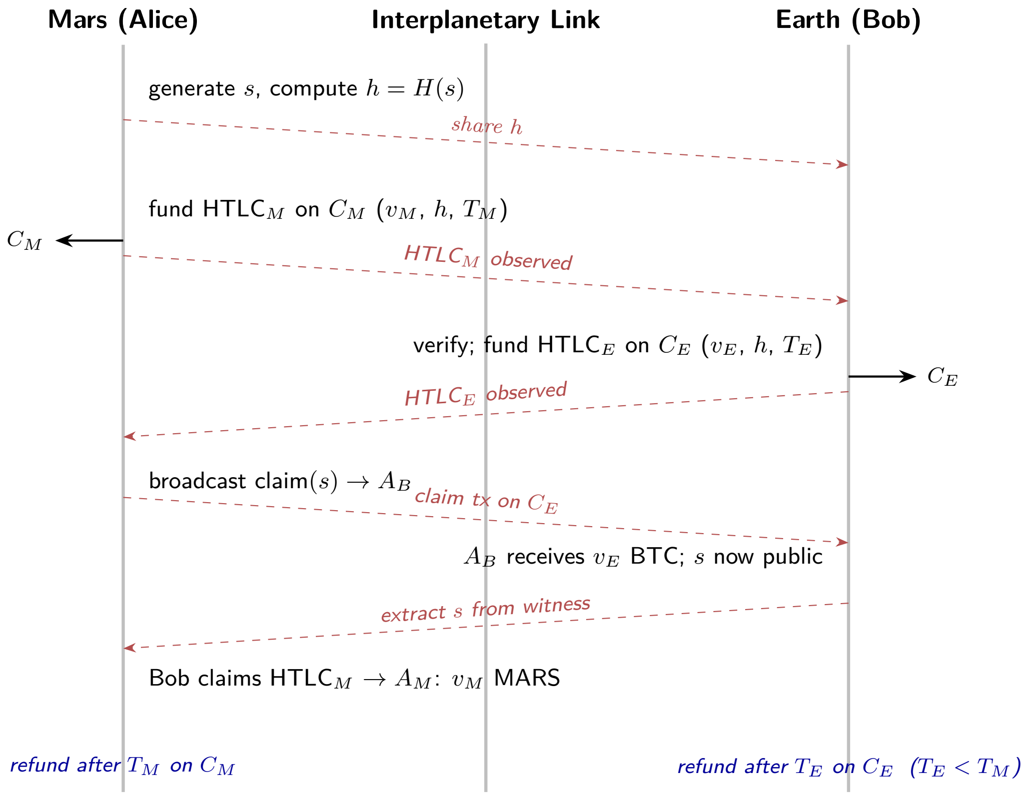

Figure 1: IPAS message-sequence diagram. Dashed red arrows are cross-planet messages subject to the light-speed delay τ_{prop}; solid black arrows are local chain transactions. No wall-clock synchronisation between planets is required. The refund-time asymmetry $T_E < T_M$ guarantees that the second funder (Earth-side Bob) can always refund before the first funder's (Mars-side Alice's) refund race opens.

where:

- $\tau_{\text{confirm},E}$ is the Earth-side confirmation buffer (e.g., 6 Bitcoin blocks) before the claim that reveals s is treated as final.
- τ_{prop} is the one-way Earth–Mars propagation delay, 3–22 minutes depending on orbital position.
- τ_{relay} is additional store-and-forward or scheduled-contact relay delay beyond light-speed.
- τ_{obs} is wallet/monitoring-worker polling latency on Mars.
- $\tau_{\text{claim},M}$ is Mars-side claim construction and broadcast time.
- $\tau_{\text{confirm},M}$ is Mars-side inclusion and confirmation buffer (e.g., 30 Marscoin blocks $\approx$ 1 hour).
- τ_{fee} is a fee-spike contingency, sized from empirical mempool congestion distributions.
- ε is a general safety margin.

This decomposition differs from Puente and Puente [2025] in two ways. First, we include an explicit τ_{relay} term distinct from τ_{prop}, because store-and-forward scheduling can contribute more delay than the physical link itself. Second, we separate τ_{fee} from ε, because fee behaviour on Bitcoin is heavy-tailed and demands its own buffer rather than being absorbed into a catch-all margin.

5.3 Numerical values by orbital regime

Table 1 gives recommended T_M and T_E values for four orbital regimes, derived by evaluating Equation 4 at each regime's τ_{prop} and adding conservative operational buffers. Marscoin's 123-second target block time is used throughout; Bitcoin's 10-minute target is used throughout.

Regime	τ_{prop} (one-way)	BTC timelock T_E	MARS timelock T_M	Safety gap Δ
Close approach	$\sim$3 min	48 h (288 blk)	96 h (2,810 blk)	48 h
Average distance	$\sim$13 min	72 h (432 blk)	144 h (4,215 blk)	72 h
Far approach	$\sim$22 min	96 h (576 blk)	192 h (5,620 blk)	96 h
Conjunction-spanning	blocked $\approx$14 d	504 h (3,024 blk)	672 h (19,670 blk)	168 h

Table 1: Recommended timelock parameters for four Earth–Mars orbital regimes. Block counts assume Bitcoin's 10-minute target block time and Marscoin's 123-second target block time. The safety gap Δ is wall-clock. The conjunction-spanning regime deliberately exceeds the blackout duration plus a one-week margin, so that neither refund activates during communication loss.

The conjunction-spanning parameters may seem excessive — nearly a month of capital lockup in the worst case — but the framing matters. For a civilisation in which a Starship cargo run takes 6–9 months, a 28-day financial settlement is ahead of the logistics chain, not behind it. The architecture matches the physics.

Figure 2 plots one-way Earth–Mars light-time across a single synodic period (approximately 780 days), with the four orbital regimes of Table 1 overlaid as horizontal safety-gap bands. The plot makes clear that close-approach parameters (48h BTC timelock) are appropriate only in a narrow window around opposition, while conjunction-spanning parameters (504h BTC timelock) are mandatory for the roughly two-week blackout at superior conjunction. Between these extremes, the 72h and 96h parameters cover the bulk of the synodic cycle.

5.4 A worked example

A Marscoin mining operator on Mars wants to purchase 100 kg of specialised carbon-fibre fabric from an Earth supplier for 0.8 BTC, paying in MARS at the prevailing IPAS quote. The current orbital regime is average distance ($\tau_{\text{prop}} \approx 13$ min). Using Table 1:

- Operator (Alice) generates s, sets T_M = current Mars tip + 4,215 blocks ($\approx$ 144 h), funds HTLC$_M$ with the agreed MARS amount and claim destination pk_{taker}.
- Taker (Bob) observes HTLC$_M$ on his Earth-side Marscoin mirror within approximately 13 minutes of propagation plus any relay scheduling. Verifies parameters, including T_M.
- Bob sets T_E = current Earth tip + 432 blocks ($\approx$ 72 h) and funds HTLC$_E$ with 0.8 BTC. Claim destination is the supplier's public key pk_{supplier}, not Alice's.
- Alice (or the supplier; see Section 4.3) claims HTLC$_E$, revealing s on C_E. The supplier's Earth wallet credits 0.8 BTC.
- Bob's Earth-side worker sees s in the claim transaction, relays to his Mars mirror, and claims HTLC$_M$. Bob's Mars wallet credits the MARS.

Typical total elapsed time: under 24 hours. Worst-case elapsed time if Bob's worker misses the reveal but still recovers before the 72-hour Earth refund: under 72 hours. At no point is any party's principal at risk beyond the agreed timelock windows, and at no point is a trusted intermediary required.

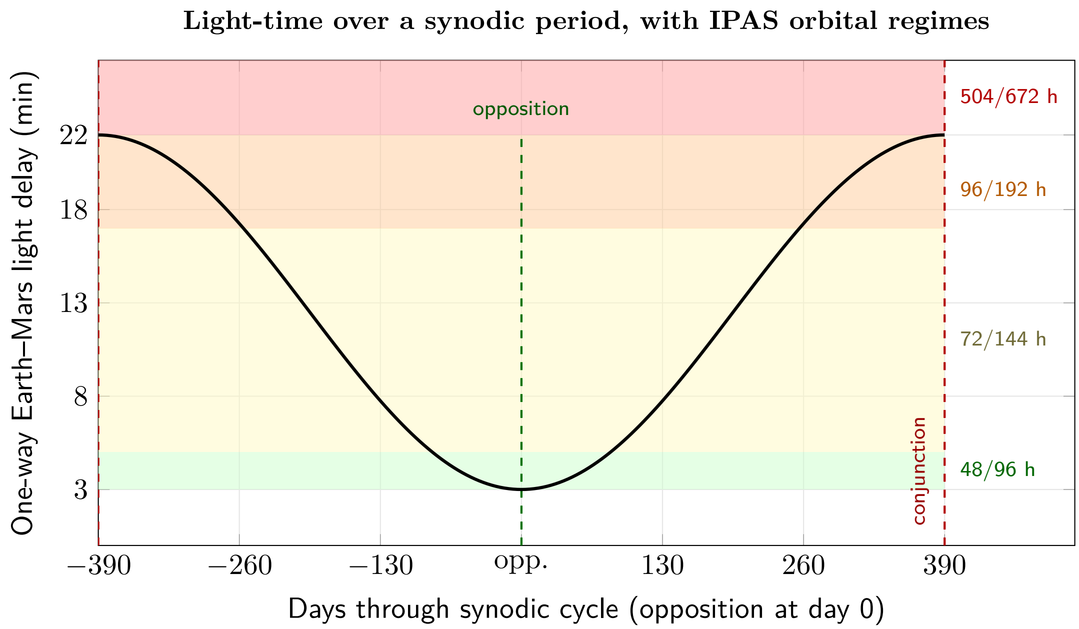

Figure 2: One-way Earth–Mars light-time across a single synodic period ($\sim$780 days), approximated as $\tau_{\text{prop}}(t) \approx 12.5 - 9.5\cos(2\pi t/780)$ minutes, overlaid with the four IPAS orbital regimes from Table 1. Dashed green marks opposition (close approach); dashed red marks conjunction at the cycle boundaries, where direct radio communication is blocked for approximately two weeks. The coloured bands indicate the orbital regime in which each parameter set is appropriate (BTC/MARS timelock in hours, right margin); an IPAS wallet can auto-select the correct regime from the current ephemeris.

6 Infrastructure: Delayed-Mirror Architecture

6.1 What each planet must observe

IPAS requires that each planet can, eventually, observe the other's chain. Neither chain depends on the other for its own consensus. Mars does not mine Bitcoin. Earth does not mine Marscoin. Each planet simply maintains a *delayed mirror* of the other's chain, sufficient to verify claim transactions and extract preimages.

6.2 A bandwidth budget

Bitcoin produces approximately 6 blocks per hour, each with a header of 80 bytes and an average full block size of approximately 1.5 MB. The minimum-viable Earth→Mars feed is block headers only:

$$B_{\text{hdr}} = 6 \times 80 \text{ bytes/h} = 480 \text{ bytes/h} \approx 4.7 \text{ KB/d}. \tag{5}$$

This is sufficient for SPV-style inclusion proofs: with header chain and a Merkle branch, a Mars wallet can verify that a specific claim transaction is included in a block without holding the full block. For unambiguous preimage extraction, full blocks are preferable:

$$B_{\text{full}} = 6 \times 1.5 \text{ MB/h} \approx 9 \text{ MB/h} \approx 216 \text{ MB/d}. \tag{6}$$

Marscoin, with 123-second target block time and small blocks, requires less still: approximately 2 MB/d for full blocks. A single-direction Earth–Mars link sized to support a human settlement — video, telemetry, science payload — carries data at rates of tens to thousands of megabits per second. Allocating 9 MB/h of that capacity to Bitcoin block relay is a rounding error on the communication budget. The infrastructure cost of IPAS at the link level is negligible.

6.3 Conjunction handling

During conjunction, no data traverses the direct Earth–Mars link. IPAS wallets must be conjunction-aware but not conjunction-dependent. Three behaviours matter:

1. **Pre-conjunction.** Wallets displaying an IPAS offer within the conjunction-window horizon automatically suggest conjunction-spanning timelocks (Table 1, row 4), or decline to construct the offer if the participant prefers to wait out the blackout.
2. **Mid-conjunction.** Both parties' wallets continue monitoring their *own* local chains. If a claim occurs on one chain during blackout, the preimage exists in that chain's data and will be discoverable when communication resumes. The extended timelocks ensure that no refund activates during the blackout window.
3. **Post-conjunction.** When communication resumes, both parties' mirrors resynchronise and pending preimage extractions complete normally.

Conjunction does not defeat IPAS. It changes its regime.

6.4 Redundant communication paths

To mitigate relay censorship, each planet should ingest the other's chain through multiple independent channels: direct deep-space radio, optical links, and store-and-forward through relay satellites or orbital infrastructure. No single relay operator should be able to deny a party's observation of a preimage reveal. This is an operational discipline, not a protocol change; IPAS benefits from the same multi-path communication philosophy NASA already applies to Mars surface assets [NASA Science, 2026].

7 IPAS vs. Interplanetary Lightning

7.1 What Lightning solves, and what it costs

Poon and Dryja [2016] introduced Lightning as an off-chain scaling mechanism. A Lightning channel is a two-of-two multisig between two participants, with HTLC updates passed off-chain to encode payment state. Scaling follows: most payments never touch the base chain. Puente and Puente [2025] proposes extending this architecture to Mars, with light-time-adjusted CLTV margins and proof-of-transit receipts for cross-planetary audit.

Lightning is a remarkable piece of engineering. It is also, eight years after launch, not the payment rail that most Bitcoin users interact with. The empirical adoption gap is not an accident; it reflects real operational costs:

- **Channel management.** Opening, closing, and rebalancing channels is a continuous operational burden. Channel liquidity must be actively managed on both sides; inbound and outbound capacity do not balance themselves.
- **Watchtower dependence.** If a counterparty broadcasts an old channel state while the user is offline, the user loses funds unless a watchtower intervenes within the dispute window.

Watchtowers are a trusted-party introduction; their business model is still unsettled.

- **Attack surface.** Harris and Zohar [2020] ("Flood & Loot"), Mizrahi and Zohar [2021] (congestion attacks), and Riard and Naumenko [2020] (time-dilation attacks) document systemic vulnerabilities that do not apply to simple on-chain HTLCs.
- **Liveness requirement.** Lightning requires participants (or their delegated watchtowers) to be online during dispute windows. The on-chain HTLC requires only that the chain itself stays online.

Every one of these costs becomes worse at interplanetary distance. Time-dilation attacks exploit delayed block propagation; 22-minute light time makes them trivially easier. Flood-and-loot attacks exploit mempool congestion during disputes; Mars-side mempools have no redundant global gossip. Watchtower liveness requires reliable inter-planet connectivity, which is precisely what conjunction removes. Channel rebalancing across the Earth–Mars link requires bidirectional liquidity that no rational operator would provide without onerous spreads.

7.2 Why on-chain HTLC wins for Mars

The on-chain IPAS construction avoids every one of these concerns:

- **No operator.** There is no Lightning service provider, no hub, no watchtower. There are two parties and two chains.
- **No channel state.** Each swap is a fresh pair of HTLCs. There is no old state that can be maliciously re-broadcast.
- **No liveness during dispute.** If a participant goes offline for a week, nothing is lost. Either the preimage has been revealed (in which case the claim can be made up to the refund height) or it has not (in which case both refunds activate).
- **Public auditability.** Every IPAS swap is fully visible on both chains. The Martian Republic treasury can publish its on-chain swap history and have it verified by any Earth-side observer with a Bitcoin node.
- **Custody sovereignty.** No third party ever holds both sides' funds. The asymmetric-timelock construction makes principal-level custodianship structurally impossible.

The argument is not that Lightning is bad. The argument is that *for Mars, simpler is safer*. An interplanetary payment system whose correctness depends on off-chain state, cooperative channel closing, watchtower liveness, and sub-dispute-window responsiveness is a payment system whose correctness depends on the absence of conjunction. The on-chain HTLC has no such dependency. A capital lockup of six days at average Earth–Mars distance buys total elimination of every Lightning operational concern.

7.3 Transparency and sovereignty

A subtler argument: on-chain HTLCs are *legible*. A researcher, a regulator, or a future Martian historian can read the BTC↔MARS exchange history directly from the two chains, with cryptographic certainty that the transactions occurred and were settled atomically. Lightning's off-chain state is, by design, opaque to non-participants. For a civilisation that will eventually need to construct institutions, norms, and economic statistics from scratch, legible settlement is a public good. IPAS is legible by default. Lightning is not.

8 Economic Implications

8.1 Interplanetary commerce

IPAS converts Marscoin from a speculative asset into a functional medium of exchange for Earth–Mars trade. A Mars colony that produces unique goods — isotopic data, rare-sample spectrometry, manufactured lunar/Martian alloys, intellectual property, tourism vouchers, governance participation rights — can sell them for Bitcoin directed to Earth-side suppliers. The colonist's MARS balance decreases; the supplier's BTC balance increases; the rate is set by the counterparties themselves, without an exchange listing, a market maker, or a regulatory intermediary.

A concrete near-term scenario: SpaceX or a successor launch provider invoices a Mars habitat for cargo manifest slots. Historically such invoices would be cleared through a US-dollar commercial account. Via IPAS, the Mars habitat can pay the launch provider in BTC directly, settling from MARS held in the habitat treasury. The launch provider never touches MARS; the habitat never touches USD. Each side operates in the medium it is natively comfortable with, and the HTLC bridges them.

8.2 Remittance

Earth-to-Mars remittance via IPAS is structurally identical to how hawala networks have operated for 1,200 years — except trustless. A parent on Earth takes a MARS offer with the child's Mars address as A_M. The parent pays BTC; the child receives MARS; the maker earns the spread. In hawala, the intermediary is *trusted*. In IPAS, the intermediary is *replaced by an HTLC*. This is what it looks like when a 1,200-year-old social institution is reimplemented as a 32-byte hash.

8.3 Capital formation

This may be the most consequential use case. Mars colonisation faces a persistent funding problem: government budgets are volatile, private investment demands return, philanthropy does not scale. IPAS introduces a new mechanism.

A Mars colony that has built economic value — scientific discoveries, mineral extraction rights, manufactured goods, IP portfolios, governance participation rights — can convert that value into Earth-side purchasing power immediately, without a bank account on Earth, without a wire service, without any terrestrial financial intermediary. It needs only a counterparty willing to trade BTC for MARS. In the reverse direction, Earth-side capital can flow to Mars almost directly: an investor, an institution, or an aid program takes IPAS offers, acquiring MARS that funds Martian governance, infrastructure, or individual colonists' accounts. The capital arrives on Mars as spendable Marscoin within the IPAS settlement window — days, not quarters.

8.4 Price discovery

IPAS offers naturally form a cross-planetary order book. The BTC/MARS exchange rate is set by makers and takers based on their own assessments. Over time, this distributed price discovery produces a market rate for interplanetary trade — the first such rate in human history, emerging from peer-to-peer transactions rather than institutional proclamation.

There is a historical parallel worth naming. In 1602, Amsterdam Bourse quotes for Dutch East India Company shares produced the world's first liquid secondary market in corporate equity; in 1848, the Chicago Board of Trade grain contracts produced the first standardised commodity futures; each time, a new asset class required a new market form. IPAS is the market form for cross-planetary value. It emerges not from a regulator's designation but from the cryptographic structure of the swap itself.

8.5 Failure case: why custodial bridges cannot work

Consider a counterfactual: a centralised BTC/MARS exchange with deposit addresses on Earth and Mars. The exchange promises 1:1 redeemability. A conjunction begins. For 14 days, no audit of the exchange's reserves is possible from either planet. If the exchange is solvent and honest, nothing happens. If it is not — if it has loaned customer deposits, if it is running fractional reserves, if its Mars-side operator has absconded — no counterparty can discover this until communication resumes. The FTX collapse of 2022 demonstrated that such concealment is not hypothetical even within a single light-speed boundary. Across conjunction, it is unsurvivable.

IPAS has no such failure mode. There is no exchange, no custody, and no reserve ratio. A pending swap either completed or will refund; no third party's solvency is on the critical path. This is not an advantage. It is the difference between a viable interplanetary financial infrastructure and an inevitable interplanetary financial fraud.

9 Mainnet Demonstration and Reference Implementation

9.1 The 12 April 2026 swap

The existence proof underlying this paper is not a thought experiment. On 12 April 2026, a maker-taker pair running the open-source Electrum-Mars wallet (v4.4.0-mars) completed a BTC↔MARS atomic swap on mainnet [Marscoin Foundation, 2026]. The parameters were:

Parameter	**Value**	**Notes**
Maker offer	104 MARS	Mars-side, funded HTLC$_M$ first
Taker payment	0.00003652 BTC	Earth-side, counter-funded HTLC$_E$
BTC timelock T_E	4 hours	$\approx$ 24 Bitcoin blocks
MARS timelock T_M	8 hours	$\approx$ 234 Marscoin blocks
Safety gap Δ	4 hours	wall-clock
Preimage	32-byte random, SHA-256	generated by maker
Monitoring (BTC)	`mempool.space` API	maker-side watcher
Monitoring (MARS)	local node	taker-side watcher
Claim path	witness-data preimage reveal	taker extracts s from BTC claim tx

Table 2: Parameters of the 12 April 2026 first mainnet BTC↔MARS atomic swap.

9.2 The 11 April refund-path test

A full end-to-end happy-path swap was not, in fact, the first on-mainnet event. On 11 April 2026 — one day prior — an earlier test stalled when a bug in cross-wallet communication of the BTC HTLC address prevented the taker's wallet from making progress. The Bitcoin HTLC was funded but the claim path was never exercised. Four hours later, the refund timelock elapsed and the Bitcoin principal returned automatically to the taker's external wallet (an Exodus wallet) via the refund script path [Marscoin Foundation, 2026].

This failed test is more valuable for this paper's argument than the happy path. It demonstrates that the *safety net is not theoretical*. Even when the swap engine's state machine cannot complete the trade, no party loses principal. The cryptographic refund path activated exactly as specified by Equation 1. This is the property that makes IPAS viable across 22-minute light delay and two-week conjunctions: the happy path is nice; the refund path is the reason the construction is safe.

9.3 Unscheduled stress: the empty-block incident

The 12 April swap was itself non-trivial. Marscoin miners happened to be producing empty blocks for approximately nine hours during the test window — a mining infrastructure issue unrelated to the swap system. Marscoin transactions sat in the mempool far longer than the expected $\sim$2 minutes per block. The Electrum-Mars swap engine, designed around a short-confirmation assumption, initially treated the delay as an expiration condition and attempted to abort. A fix was applied mid-test to relax the abort heuristic; the swap completed successfully once the Marscoin transaction finally confirmed [Marscoin Foundation, 2026].

This is exactly the class of operational hazard that scales into the interplanetary regime. The engineering response — widen the confirmation buffer, distinguish network delay from contract expiry, and decouple the worker's state transitions from wall-clock assumptions — is precisely the discipline captured by the τ_{confirm} and τ_{fee} terms of Equation 4. The Earth-based implementation already encountered, at small scale, the failure mode that interplanetary deployment will encounter at large scale. Solving it on Earth, in public, with small amounts of money, is the correct way to earn the right to deploy it between planets.

9.4 Reference implementation

The full reference implementation is open source at `github.com/marscoin/electrum-mars`; the atomic-swap release is tagged `v4.4.0-mars` (commit `cff92e1`) [Lopin, 2026a]. Salient architectural features:

- **Background SwapWorker.** A dedicated worker drives the HTLC state machine automatically once a swap is initiated, surviving wallet restarts and polling the counterparty chain at fixed intervals.
- **Order book via ElectrumX relay.** Offers propagate through the existing Marscoin ElectrumX gossip network, with no new protocol added.
- **Auto-Maker.** A passive market-making mode connects to a price feed, applies a configurable spread, and automatically posts and manages offers. Any wallet can become a miniature decentralised exchange.
- **Planned Taproot migration.** A future migration to Taproot-based adaptor signatures would hide the HTLC mechanics inside a standard-looking transaction, improving efficiency and privacy.

IPAS requires no change to this implementation beyond parameter selection. Extending the timelocks and adding conjunction-aware UI is a superset of the current v4.4.0-mars release, not a rewrite.

10 Security, Failure Modes, and Limitations

IPAS inherits the security of its constituent HTLCs and the underlying chains. It adds no new cryptographic assumption. It does, however, surface several operational concerns that must be managed.

Monitoring failure. If a participant's worker fails to observe a preimage reveal in time, the swap completes against them — the counterparty claims, and the timelock refund closes without recovery. The Mars-side operator of a MARS-holding maker must therefore run redundant monitoring. In practice this is a low-cost operational discipline; the Marscoin wallet's swap engine already implements it for terrestrial swaps [Lopin, 2026b].

Fee spikes and inclusion races. Bitcoin's fee market is heavy-tailed. A claim transaction broadcast "just before timeout" may not confirm in time. IPAS implementations must reserve fee-spike buffer (captured in τ_{fee} in Equation 4) and support child-pays-for-parent re-broadcasting. This is an extant HTLC problem, not an IPAS-specific one.

Reorg risk. Treating a redemption as final before adequate confirmation can cause a counterparty to claim against a later-reorged chain state. This is handled by the τ_{confirm} terms in Equation 4; six confirmations on Bitcoin and thirty on Marscoin are conservative defaults that push reorg probability well below 10^{-6} [Nakamoto, 2008].

Griefing and liquidity lock-up. A malicious counterparty can fund one leg and then decline to reveal the preimage, forcing both sides to wait out the timelock for a refund. This is an old HTLC cost, but at interplanetary distance it scales with T_M — up to 28 days of locked capital in conjunction-spanning mode. Bridge liquidity on this scale will likely be priced accordingly; spreads will reflect the capital cost of time.

Censorship. Neither IPAS nor any other cryptographic construction prevents relay censorship, miner censorship, or state intervention. Multi-path communication (Section 6) mitigates relay-level censorship; open relay networks and multiple mining pools mitigate miner-level censorship. State-level censorship is a political problem, not a protocol problem.

Off-chain performance. IPAS settles payment atomically. It does not adjudicate whether the physical cargo arrived on Mars, whether the scientific dataset met its specification, whether the patent licence was lawful in both jurisdictions. These are off-chain disputes requiring contract law, arbitration, or reputation systems. IPAS is the settlement primitive; it is not the entire commercial apparatus.

Scope. This paper does not provide: a formal proof of optimal timeout policy, a full implementation specification of the offer-propagation format, or a simulation of IPAS under realistic Deep Space Network contact schedules. These are natural follow-ups and are planned for the Marscoin technical report series.

11 Political Economy

11.1 Compatibility with multiple sovereignty regimes

IPAS does not prescribe a political settlement for Earth–Mars monetary relations. It is compatible with at least three regimes:

1. **Mars-local currency.** Mars maintains Marscoin as its domestic medium and uses IPAS selectively for cross-planet trade, remittance, and capital flow. This is the position of the Marscoin literature since 2014 [Lopin et al., 2014].
2. **Pegged or federated architecture.** Mars uses a locally-issued token representing a peg to Bitcoin or another Earth monetary base, with IPAS bridging the peg. This preserves trust-minimisation at the exchange layer even under federated issuance.
3. **Restricted interchange.** A sovereign-Mars regime [Haqq-Misra, 2024] limits or prohibits Earth–Mars currency flow. IPAS remains technically relevant but politically bounded: the mechanism exists, its use is policy-constrained.

The cryptographic mechanism does not settle the sovereignty question. It changes what becomes technically feasible under whichever regime is chosen.

11.2 Why local chains won

The 2013–2014 Marscoin argument for a Mars-local chain rested on three observations, all of which have aged well. First, light-speed delay makes synchronous global consensus across Earth and Mars unviable; any protocol that requires it will fail on planetary separation. Second, a local chain allows a Mars settlement to clear its own economic life without depending on continuous connectivity to Earth. Third, cross-planet transfer is a *less frequent, higher-value* operation and deserves a distinct settlement interface from everyday in-domain transactions.

The contemporaneous "Interplanetary Coin" third-chain proposal (Section 1) attempted to supply that interface by introducing a new chain running at a slow, delay-tolerant cadence. In retrospect, this was the less economical answer. A third chain would have required its own consensus, its own validator set, its own token economics, and its own governance — all to solve a problem that the bilateral HTLC solves with no new infrastructure at all. The correct architecture is two sovereign local chains plus a cryptographic condition that binds them only when value needs to move between them. That is what IPAS is.

11.3 Content-addressed settlement as a general principle

The analogy to IPFS [Benet, 2014] is more than a naming convention. IPFS retrieves content by hash rather than by server location; IPAS settles value by hash rather than by synchronous ledger state. What travels between Earth and Mars in an IPAS swap is not consensus, not shared state, not continuous mutual observation. It is *evidence*: delayed knowledge that a preimage satisfying a public hash commitment has been revealed on the remote chain. The economic link between domains is keyed to a hash-identified condition that verifies or fails to verify locally.

This is a general design principle for interplanetary infrastructure. Whenever a task can be accomplished by verifiable delayed evidence rather than continuous synchronous coherence, the former is strictly more robust. IPFS showed this for data; IPAS shows it for money. Future interplanetary primitives — for naming, for identity, for delivery attestation — will benefit from the same discipline.

12 Conclusion

The central result of this paper is narrow and, we believe, defensible: the on-chain Hash Time-Locked Contract is the correct settlement primitive for Earth–Mars commerce. The construction has been proven in production between Bitcoin and Marscoin on Earth; the adaptation required for interplanetary use is parameter selection, not protocol redesign. The cryptographic atomicity of the HTLC is distance-agnostic. The operational safety of the HTLC is distance-sensitive but bounded by engineering quantities that we have made explicit: propagation delay, relay scheduling, confirmation buffers, fee contingency, and conjunction policy. Equation 4 and Table 1 give concrete parameter ranges for each orbital regime.

Three secondary claims follow. First, the 2013 Marscoin discussion of cross-planet value transfer — including the (unbuilt) "Interplanetary Coin" third-chain proposal — posed the correct problem but reached for an unnecessarily complex mechanism. A bilateral HTLC between two sovereign chains, published in the same year on BitcoinTalk [Nolan, 2013], is the simpler and stronger construction. IPAS makes that connection explicit. Second, the on-chain HTLC is preferable to Lightning-based interplanetary proposals [Puente and Puente, 2025] because it eliminates operator, channel-state, and watchtower assumptions whose failure modes are amplified by light-speed delay and conjunction. Third, the destination-address construction (Section 4.3) converts the atomic swap from a trading primitive into a commerce primitive without introducing a third trusted role, and this is what enables the concrete use cases — remittance, merchant settlement, sovereign treasury funding, scientific-data markets — developed in Section 8.

Significant work remains. Timeout policy under realistic Deep Space Network contact schedules warrants simulation. Offer propagation across the interplanetary link needs specification beyond the current ElectrumX gossip mechanism. A planned Taproot-adaptor-signature migration [Lopin, 2026a] will improve efficiency and privacy but adds engineering complexity that must be validated on Earth before interplanetary deployment. And, at a longer horizon, a bootstrapping payload for a future Mars settlement — containing the Marscoin node software, the Electrum-Mars wallet, governance tooling, and the IPAS swap engine — is a natural integration artifact for the Marscoin Foundation's roadmap, though its design is beyond the scope of this paper.

The contribution here is therefore modest in claim and specific in content: a working cryptographic primitive, a parameter regime for a known physical link, an engineering inequality, a mainnet existence proof, and a comparison against the nearest adjacent proposal. None of these elements is individually novel. Their combination, as far as we have been able to establish, has not previously been assembled into a single Mars-oriented settlement architecture.

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